Kinematic Calibration of a Pan-Tilt Mounted Planar LiDAR System

Mathematical Formulation, Implementation, and Experimental Results

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Abstract

This report presents a comprehensive kinematic calibration methodology for a pan-tilt unit (PTU) carrying a Hokuyo URG-04LX planar LiDAR sensor. The system sweeps the 2D laser across multiple pan-tilt configurations while facing a flat wall, and estimates 14 kinematic correction parameters by minimizing the global plane-consistency cost of the projected 3D points. We detail the complete mathematical formulation of every stage of the pipeline: grid-scan data acquisition with RANSAC line fitting, per-capture line quality filtering, forward kinematics with correction parameterization, least-squares optimization with Tikhonov regularization, observability analysis via numerical SVD, and cross-validation. Experimental results on 31 scan configurations demonstrate a 12.3% reduction in global plane RMSE after calibration, with the per-capture line quality filter alone accounting for a 40% noise reduction. We identify fundamental observability limitations of single-wall calibration and discuss strategies for achieving higher accuracy.

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1 Introduction

A planar LiDAR sensor mounted on a pan-tilt mechanism can generate three-dimensional point clouds by sweeping the 2D scan plane through a range of orientations. The accuracy of the resulting 3D reconstruction depends critically on the precise knowledge of the kinematic chain relating the laser frame to a fixed base frame. Nominal values from hand measurements and mechanical drawings are typically insufficient for applications requiring sub-centimetre accuracy.

Kinematic calibration estimates the true geometric parameters of the mechanism—joint origins, orientations, and encoder offsets—from observed data. The fundamental idea is to view a geometrically known target (a flat wall) from multiple configurations and solve for the parameters that best explain the observations.

1.1 System Overview

Our system consists of:

- **LiDAR:** Hokuyo URG-04LX, planar scanner, 240° FOV, ∼60 mm accuracy specification.
- **Pan-Tilt Unit:** Two-axis servo mechanism controlled via Arduino, reporting joint states via ROS `/joint_states`.
- **Software:** ROS 1 Noetic, Python 3 calibration pipeline.

1.2 Report Outline

The remainder of this report is organized as follows: Section 2 describes the nominal kinematic chain from the URDF and the correction parameterization. Section 3 details the data acquisition procedure including grid generation, FOV extraction, and RANSAC line fitting. Section 4 presents the per-capture line quality filter. Section 5 develops the optimization formulation, cost function, and solver configuration. Section 6 covers the observability analysis. Section 7 describes the validation methodology. Section 8 presents the experimental results. Section 9 discusses limitations and future work needed for improved calibration.

2 Kinematic Chain and Correction Parameterization

2.1 URDF Kinematic Chain

The mechanism is described by the following URDF link–joint chain:

$$\texttt{base_link} \xrightarrow{\text{joint2 (pan)}} \texttt{pan_link} \xrightarrow{\text{joint1 (tilt)}} \texttt{tilt_link} \xrightarrow{\text{hokuyo_base_joint}} \texttt{hokuyo_base} \xrightarrow{\text{hokuyo_joint}} \texttt{laser}$$

Each joint has a fixed origin transform $\mathbf{T}_{\text{origin}} \in \mathrm{SE}(3)$ specified by a translation vector $\mathbf{t} = (x, y, z)$ and an orientation given by roll-pitch-yaw angles (ϕ, θ, ψ).

2.1.1 Homogeneous Transform from URDF Parameters

Given $\mathbf{t} = (x, y, z)$ and $\mathrm{rpy} = (\phi, \theta, \psi)$, the URDF convention constructs the 4×4 homogeneous transformation matrix as:

$$\mathbf{T}(\mathbf{t}, \mathrm{rpy}) = \begin{bmatrix} \mathbf{R}_z(\psi)\,\mathbf{R}_y(\theta)\,\mathbf{R}_x(\phi) & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \tag{1}$$

where $\mathbf{R}_x(\phi)$, $\mathbf{R}_y(\theta)$, $\mathbf{R}_z(\psi)$ are the elementary rotation matrices about the x, y, z axes respectively.

Critical convention: The product $\mathbf{R}_z(\psi)\,\mathbf{R}_y(\theta)\,\mathbf{R}_x(\phi)$ corresponds to *extrinsic* X-Y-Z rotations (equivalently, *intrinsic* Z-Y'-X''). In the `scipy` library, this is specified as `Rotation.from_euler('XYZ', rpy)`, where the uppercase letters denote the extrinsic (fixed-axis) convention.

> **Bug found during this project:** The original code used `'xyz'` (lowercase, intrinsic), which applies rotations in the opposite order. Correcting to `'XYZ'` (extrinsic) reduced the uncalibrated plane RMSE from 56 mm to 27 mm—a single-character fix with enormous impact.

2.1.2 Nominal Transform Values

The nominal (hand-measured) URDF parameters are:

Table 1: Nominal URDF joint parameters.

Joint	Frames	xyz (m)	rpy (rad)	Axis
`joint2` (pan)	base → pan	(0, −0.0285, 0.092)	(0, 0, 0)	(0, 0, −1)
`joint1` (tilt)	pan → tilt	(0, −0.009, 0.022)	(0, 1.57, 0)	(0, 0, 1)
`hokuyo_base`	tilt → hokuyo_base	(−0.045, 0, 0)	(0, 0, 0)	—
`hokuyo_joint`	hokuyo_base → laser	(−0.061, 0.014, 0.040)	(1.57, 3.14, −1.57)	—

In our implementation, the two fixed joints (`hokuyo_base_joint` and `hokuyo_joint`) are composed into a single transform:

$$\mathbf{T}^{\text{nom}}_{\text{laser}} = \mathbf{T}_{\text{hbase}} \cdot \mathbf{T}_{\text{hjoint}} \tag{2}$$

The nominal static transforms used in forward kinematics are therefore:

$$\mathbf{T}^{\text{nom}}_{\text{pan}} = \mathbf{T}\Big((0,\ -0.0285,\ 0.092),\ (0,\ 0,\ 0)\Big) \tag{3}$$

$$\mathbf{T}^{\text{nom}}_{\text{tilt}} = \mathbf{T}\Big((0,\ -0.009,\ 0.022),\ (0,\ 1.57,\ 0)\Big) \tag{4}$$

$$\mathbf{T}^{\text{nom}}_{\text{laser}} = \mathbf{T}\Big((-0.045,\ 0,\ 0),\ (0,\ 0,\ 0)\Big) \cdot \mathbf{T}\Big((-0.061,\ 0.014,\ 0.040),\ (1.57,\ 3.14,\ -1.57)\Big) \tag{5}$$

2.2 Correction Parameterization

We parameterize kinematic errors as small corrections applied to the nominal transforms. Consider a nominal transform $\mathbf{T}^{\text{nom}}$ and a small perturbation consisting of a translation $\delta\mathbf{t} \in \mathbb{R}^3$ and a rotation $\delta\mathbf{r} = (\delta\phi, \delta\theta, \delta\psi) \in \mathbb{R}^3$:

$$\mathbf{T}^{\text{corr}} = \mathbf{T}^{\text{nom}} \cdot \Delta\mathbf{T}, \qquad \Delta\mathbf{T} = \begin{bmatrix} \mathbf{R}(\delta\mathbf{r}) & \delta\mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix} \tag{6}$$

where $\mathbf{R}(\delta\mathbf{r}) = \mathbf{R}_z(\delta\psi)\,\mathbf{R}_y(\delta\theta)\,\mathbf{R}_x(\delta\phi)$ follows the same extrinsic XYZ convention. This right-multiplication means the correction is expressed in the *child* frame of the joint, which is the natural choice for body-frame perturbations.

2.3 The 14-Parameter Model

For calibration with a single flat wall, the tilt joint origin corrections are *degenerate* with the laser mount corrections (see Section 6). We therefore fix the tilt joint geometry at its nominal values and define a 14-parameter correction vector:

$$\boldsymbol{\delta} = \Big[\underbrace{\delta\mathbf{t}_{\text{pan}},\ \delta\mathbf{r}_{\text{pan}}}_{\text{6: pan origin}},\ \underbrace{\varepsilon_{\text{pan}},\ \varepsilon_{\text{tilt}}}_{\text{2: encoder zeros}},\ \underbrace{\delta\mathbf{t}_{\text{laser}},\ \delta\mathbf{r}_{\text{laser}}}_{\text{6: laser mount}}\Big] \in \mathbb{R}^{14} \tag{7}$$

The parameters are:

1. $\delta\mathbf{t}_{\text{pan}} = (\delta t_x, \delta t_y, \delta t_z)$: Translation correction to the pan joint origin.
2. $\delta\mathbf{r}_{\text{pan}} = (\delta\phi, \delta\theta, \delta\psi)$: Rotation correction to the pan joint origin.
3. ε_{pan}: Pan encoder zero offset (rad). The true pan angle is $\theta_{\text{pan}} + \varepsilon_{\text{pan}}$.
4. $\varepsilon_{\text{tilt}}$: Tilt encoder zero offset (rad). The true tilt angle is $\theta_{\text{tilt}} + \varepsilon_{\text{tilt}}$.
5. $\delta\mathbf{t}_{\text{laser}}$: Translation correction to the composite laser transform.
6. $\delta\mathbf{r}_{\text{laser}}$: Rotation correction to the composite laser transform.

2.4 Forward Kinematics with Corrections

Given joint angles (θ_p, θ_t) and correction vector $\boldsymbol{\delta}$, the forward kinematics computes the laser-to-base transform:

$$\boxed{\mathbf{T}^{\text{base}}_{\text{laser}}(\theta_p, \theta_t;\ \boldsymbol{\delta}) = \mathbf{T}^{\text{corr}}_{\text{pan}} \cdot \mathbf{R}_z\Big(-(\theta_p + \varepsilon_{\text{pan}})\Big) \cdot \mathbf{T}^{\text{nom}}_{\text{tilt}} \cdot \mathbf{R}_z\Big(\theta_t + \varepsilon_{\text{tilt}}\Big) \cdot \mathbf{T}^{\text{corr}}_{\text{laser}}} \tag{8}$$

where:

$$\mathbf{T}^{\text{corr}}_{\text{pan}} = \mathbf{T}^{\text{nom}}_{\text{pan}} \cdot \Delta\mathbf{T}(\delta\mathbf{t}_{\text{pan}}, \delta\mathbf{r}_{\text{pan}}) \tag{9}$$

$$\mathbf{T}^{\text{corr}}_{\text{laser}} = \mathbf{T}^{\text{nom}}_{\text{laser}} \cdot \Delta\mathbf{T}(\delta\mathbf{t}_{\text{laser}}, \delta\mathbf{r}_{\text{laser}}) \tag{10}$$

and $\mathbf{R}_z(\alpha)$ is the 4×4 homogeneous rotation about z:

$$\mathbf{R}_z(\alpha) = \begin{bmatrix} \cos\alpha & -\sin\alpha & 0 & 0 \\ \sin\alpha & \cos\alpha & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{11}$$

Joint axis conventions:

- The pan joint (joint2) has axis (0, 0, −1), so the rotation is $\mathbf{R}_z(-\theta)$ rather than $\mathbf{R}_z(\theta)$.
- The tilt joint (joint1) has axis (0, 0, 1). Its origin includes an RPY of (0, 1.57, 0) which rotates the local z-axis to effectively create tilt motion.

2.5 Point Projection

A 2D laser point $\mathbf{p}_L = (x_L,\ y_L)$ in the laser frame is projected to the base frame via:

$$\mathbf{p}_{\text{base}} = \mathbf{T}^{\text{base}}_{\text{laser}} \cdot \begin{bmatrix} x_L \\ y_L \\ 0 \\ 1 \end{bmatrix} \tag{12}$$

The third coordinate is zero because the planar LiDAR scans in the xy-plane of its own frame.

3 Data Acquisition

3.1 Grid-Scan Strategy

We sample the joint space on a regular grid:

$$\mathcal{G} = \left\{ (\theta_p^{(i)}, \theta_t^{(j)}) : \theta_p^{(i)} \in \text{linspace}(-\Theta_p, +\Theta_p, N_p),\ \theta_t^{(j)} \in \text{linspace}(-\Theta_t, +\Theta_t, N_t) \right\} \tag{13}$$

where Θ_p and Θ_t are the pan and tilt limits (typically 25° and 15°), and $N_p = N_t = 5$ gives a $5 \times 5 = 25$ configuration grid (expanded to $5 \times 7 = 35$ in practice with intermediate tilt values).

At each grid pose $(\theta_p^{(k)}, \theta_t^{(k)})$:

1. The PTU is commanded to the target angles via `/joint_command` at 50 Hz.
2. Convergence is verified by checking $|\theta_{\text{current}} - \theta_{\text{target}}| < \epsilon$ where $\epsilon = 1°$.
3. After convergence, the system waits a settle time $t_s = 1.5\,\text{s}$.
4. A single `/scan` message is captured.
5. The actual encoder-reported joint angles are recorded from `/joint_states`.

3.2 FOV Extraction

The Hokuyo URG-04LX has a 240° field of view, but the calibration wall subtends only a small angular region near the forward direction. We extract a region of interest (ROI) by keeping only scan points whose bearing angle satisfies:

$$\tilde{P}_k = \left\{ \mathbf{p} \in P_k : |\alpha(\mathbf{p})| < \alpha_{\max} \right\} \tag{14}$$

where $\alpha(\mathbf{p})$ is the bearing of point $\mathbf{p}$ in the laser frame and $\alpha_{\max}$ is the half-FOV parameter (typically 10–15°). Only points within the valid range interval $[r_{\min}, r_{\max}]$ are retained.

Each valid range measurement r_i at angle $\alpha_i = \alpha_{\min} + i \cdot \Delta\alpha$ is converted to Cartesian coordinates:

$$\mathbf{p}_i = \begin{pmatrix} r_i \cos \alpha_i \\ r_i \sin \alpha_i \end{pmatrix} \tag{15}$$

3.3 RANSAC Line Fitting

Since the laser is viewing a flat wall, the extracted 2D points should lie on a straight line. We use RANSAC (Random Sample Consensus) to robustly fit this line and reject outliers.

Algorithm 1 RANSAC 2D Line Fit

Require: Point set $\tilde{P} = \{\mathbf{p}_1, \ldots, \mathbf{p}_N\} \subset \mathbb{R}^2$, threshold τ, iterations K
Ensure: Line direction $\hat{\mathbf{v}}$, mean $\bar{\mathbf{p}}$, inlier set $\mathcal{I}$

```
n_best ← 0
for k = 1, ..., K do                                   ▷ K = 600
    Sample two distinct points p_a, p_b uniformly at random
    v ← (p_b − p_a) / ‖p_b − p_a‖
    n ← (−v_y, v_x)                                    ▷ Line normal (perpendicular)
    d_i ← |(p_i − p_a)^⊤ n| for each p_i
    n_inliers ← |{i : d_i ≤ τ}|                        ▷ τ = 15 mm
    if n_inliers > n_best then
        Update best model
    end if
    if n_inliers > 0.7 · N then
        break                                          ▷ Early termination
    end if
end for
Collect inliers I for the best model
Refine: compute p̄ = mean(I), centre inliers, compute SVD to get refined direction
    v̂ as the first right singular vector of the centred inlier matrix
```

The SVD refinement step computes the best-fit line through the inliers in a least-squares sense. Given the $M \times 2$ centred inlier matrix $\mathbf{C} = \mathcal{I} - \mathbf{1}\bar{\mathbf{p}}^\top$, we compute $\mathbf{C} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top$ and take $\hat{\mathbf{v}} = \mathbf{V}_{:,1}$ (the direction of maximum variance).

Each capture stores: inlier points $\mathcal{I}_k$, line direction $\hat{\mathbf{v}}_k$, line mean $\bar{\mathbf{p}}_k$, pan angle $\theta_p^{(k)}$, tilt angle $\theta_t^{(k)}$, and timestamp.

3.4 Data Storage

All captures are saved to a compressed NumPy archive (`.npz`) containing:

- `n_captures`: number of successful captures
- `fov_half_deg`: the FOV parameter used
- For each capture k: `cap_k_pan_rad`, `cap_k_tilt_rad`, `cap_k_line_dir`, `cap_k_line_mean`, `cap_k_inlier_points`, `cap_k_all_points`, `cap_k_timestamp`

4 Per-Capture Line Quality Filter

Before the calibration solver processes the data, we apply a quality filter based on the following observation:

> **Principle:** If a single scan's inlier points do not conform to a straight line with sufficient precision, those points cannot reliably contribute to a global plane fit. Since the Hokuyo's scan plane intersects a flat wall in a line, the per-capture line RMSE is a lower bound on the achievable 3D reconstruction accuracy.

4.1 Line RMSE Computation

For capture k with inlier points $\mathcal{I}_k$, fitted line direction $\hat{\mathbf{v}}_k$, and line mean $\bar{\mathbf{p}}_k$, the perpendicular distance of each point to the line is:

$$d_i = (\mathbf{p}_i - \bar{\mathbf{p}}_k)^\top \mathbf{n}_\perp, \qquad \mathbf{n}_\perp = \begin{pmatrix} -\hat{v}_y \\ \hat{v}_x \end{pmatrix} \tag{16}$$

where $\mathbf{n}_\perp$ is the unit normal perpendicular to the line direction.

The line RMSE for capture k is:

$$\mathrm{RMSE}_{\mathrm{line}}^{(k)} = \sqrt{\frac{1}{|\mathcal{I}_k|} \sum_{i \in \mathcal{I}_k} d_i^2} \tag{17}$$

4.2 Two-Stage Filtering

The filter operates in two stages:

Stage 1: Capture Rejection. A capture is rejected entirely if its line RMSE exceeds a threshold τ_{line}:

$$\mathrm{RMSE}_{\mathrm{line}}^{(k)} > \tau_{\mathrm{line}} \implies \text{reject capture } k \tag{18}$$

The default threshold is $\tau_{\mathrm{line}} = 3.0\,\mathrm{mm}$.

Stage 2: Point Clipping. Within accepted captures, individual points with excessive perpendicular error are removed:

$$|d_i| \cdot 1000 > \tau_{\mathrm{pt}} \implies \text{remove point } i \tag{19}$$

where $\tau_{\mathrm{pt}} = 2 \cdot \tau_{\mathrm{line}}$ by default (6.0 mm).

4.3 Empirical Impact

In our dataset of 31 captures, applying the 3.0 mm line quality filter:

- Rejected 12 captures (mostly at extreme pan angles $\pm 25°$ and $+15°$), retaining 19.
- Clipped 12 individual outlier points from the remaining captures.
- Retained 1012 of 1612 original points.
- **Reduced uncalibrated plane RMSE from 26.9 mm to 16.2 mm** —a 40% improvement from filtering alone.

The captures at extreme pan angles exhibit higher noise because the wall is seen at a shallow angle, increasing the range uncertainty's effect on the reconstructed point position.

5 Calibration Solver

5.1 Point Subsampling

To control computational cost, inlier points within each capture are deterministically subsampled:

$$\mathrm{step} = \max\left(1,\ \left\lfloor \frac{|\mathcal{I}_k|}{M_{\max}} \right\rfloor\right), \qquad \mathcal{I}_k^{\mathrm{sub}} = \{\mathbf{p}_0,\ \mathbf{p}_{\mathrm{step}},\ \mathbf{p}_{2\cdot\mathrm{step}},\ \ldots\} \tag{20}$$

where $M_{\max} = 30$ points per capture. This ensures uniform spatial coverage within each scan while keeping the total point count manageable.

5.2 Initial Plane Estimation

Before optimization, we estimate the wall plane from uncalibrated forward kinematics ($\boldsymbol{\delta} = \mathbf{0}$). All subsampled points are projected to the base frame and stacked into a matrix $\mathbf{P} \in \mathbb{R}^{M\times 3}$. The plane is fitted via SVD:

1. Compute the centroid: $\bar{\mathbf{p}} = \frac{1}{M}\sum_{i=1}^{M} \mathbf{p}_i$
2. Centre the points: $\tilde{\mathbf{P}} = \mathbf{P} - \mathbf{1}\bar{\mathbf{p}}^\top$
3. Compute the SVD: $\tilde{\mathbf{P}} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$
4. The plane normal is the third right singular vector: $\hat{\mathbf{n}} = \mathbf{V}_{:,3}$ (direction of minimum variance)
5. The plane offset is: $d = \hat{\mathbf{n}}^\top \bar{\mathbf{p}}$

This yields the initial plane $(\hat{\mathbf{n}}_0, d_0)$ used to seed the optimizer. Its sign is chosen so that $d_0 > 0$ (wall is in front of the sensor).

5.3 Optimization Variables

The full optimization variable is:

$$\mathbf{x} = \Big[\underbrace{\boldsymbol{\delta}}_{\text{14 kinematic}}, \underbrace{\mathbf{n},\ d}_{\text{4 plane}}\Big] \in \mathbb{R}^{18} \tag{21}$$

The plane parameters $(\mathbf{n}, d)$ are co-optimized because the "true" wall plane is unknown and must be estimated simultaneously with the kinematic corrections. This avoids the chicken-and-egg problem of needing the plane to evaluate the cost, but needing good kinematics to estimate the plane.

5.4 Cost Function

The cost function is a sum of residuals $\mathbf{r}(\mathbf{x}) = [r_1, \ldots, r_N]$ minimized in the least-squares sense $\min_{\mathbf{x}} \frac{1}{2}\|\mathbf{r}(\mathbf{x})\|^2$. The residual vector consists of three components:

5.4.1 (A) Global Plane Consistency

For each capture k and each subsampled point $\mathbf{p}_j \in \mathcal{I}_k^{\text{sub}}$:

$$r_{k,j}^{\text{plane}} = \alpha \cdot \Big(\hat{\mathbf{n}}^\top \mathbf{p}_j^{\text{base}} - d\Big) \tag{22}$$

where $\mathbf{p}_j^{\text{base}} = \mathbf{T}_{\text{laser}}^{\text{base}}(\theta_p^{(k)}, \theta_t^{(k)}; \boldsymbol{\delta}) \cdot (\mathbf{p}_j^L, 0, 1)^\top$ is the point projected to the base frame, $\hat{\mathbf{n}} = \mathbf{n}/\|\mathbf{n}\|$ is the unit normal, and $\alpha = 5.0$ is the plane weight.

The quantity $\hat{\mathbf{n}}^\top \mathbf{p} - d$ is the *signed distance* from point $\mathbf{p}$ to the plane $\{\mathbf{x} : \hat{\mathbf{n}}^\top \mathbf{x} = d\}$. For a perfect calibration and a truly flat wall, all these residuals would be zero.

5.4.2 (B) Normal Unit-Length Constraint

Since $\mathbf{n}$ is optimized as a free 3-vector, we add a soft constraint to keep it unit-length:

$$r^{\text{norm}} = \omega_n \cdot \Big(\|\mathbf{n}\| - 1\Big), \qquad \omega_n = 10 \tag{23}$$

This penalty drives $\|\mathbf{n}\| \to 1$ without requiring an explicit constraint, which would complicate the least-squares formulation.

5.4.3 (C) Tikhonov Regularization

To prevent overfitting to noise (particularly for poorly observed parameters), we add an ℓ_2 penalty on the correction vector:

$$r_i^{\text{reg}} = \sqrt{\lambda} \cdot \delta_i, \qquad i = 1, \ldots, 14 \tag{24}$$

where λ is the regularization weight. This is equivalent to the Tikhonov regularization term $\lambda \|\boldsymbol{\delta}\|^2$ in the squared cost, which penalizes large deviations from the nominal geometry and implicitly encodes the prior that the hand-measured URDF is approximately correct.

5.4.4 Complete Residual Vector

The full residual vector has $N_{\text{pts}} + 1 + 14$ elements:

$$\mathbf{r}(\mathbf{x}) = \begin{bmatrix} \alpha \cdot (\hat{\mathbf{n}}^\top \mathbf{p}_1^{\text{base}} - d) \\ \vdots \\ \alpha \cdot (\hat{\mathbf{n}}^\top \mathbf{p}_{N_{\text{pts}}}^{\text{base}} - d) \\ 10 \cdot (\|\mathbf{n}\| - 1) \\ \sqrt{\lambda} \cdot \delta_1 \\ \vdots \\ \sqrt{\lambda} \cdot \delta_{14} \end{bmatrix} \tag{25}$$

5.5 Solver Configuration

The optimization problem is solved using `scipy.optimize.least_squares` with the Trust Region Reflective (`trf`) method, which supports box constraints. The bounds are:

$$\begin{aligned} |\delta t_i| &\le B_t && \text{(translation bound, default 5 mm)} \\ |\delta r_i| &\le B_r && \text{(rotation bound, default 0.05 rad} \approx 2.87^\circ) \\ |\varepsilon_{\text{enc}}| &\le B_e && \text{(encoder bound, default } 10^\circ) \\ |n_i| \le 1, \quad |d| &\le 20 && \text{(plane parameters)} \end{aligned} \tag{26}$$

The solver uses very tight convergence criteria ($\texttt{ftol} = \texttt{xtol} = \texttt{gtol} = 10^{-12}$) to ensure the solution is driven by bound constraints or the cost landscape rather than premature termination.

5.6 Output

The solver returns:

- The optimal correction vector $\boldsymbol{\delta}^* \in \mathbb{R}^{14}$.
- Encoder zero offsets $(\varepsilon_{\text{pan}}^*, \varepsilon_{\text{tilt}}^*)$.
- Corrected URDF joint origins computed by applying $\boldsymbol{\delta}^*$ to the nominal transforms via eq. (6).

6 Observability Analysis

Not all 14 kinematic parameters may be identifiable from the available data. The *observability analysis* determines which parameters (or linear combinations thereof) can actually be estimated.

6.1 Numerical Jacobian

We compute the Jacobian of the residual vector with respect to the 14 kinematic parameters using finite differences:

$$J_{i,j} = \frac{r_i(\mathbf{x} + \epsilon \mathbf{e}_j) - r_i(\mathbf{x})}{\epsilon}, \qquad j = 1, \ldots, 14 \tag{27}$$

where $\epsilon = 10^{-7}$ and $\mathbf{e}_j$ is the j-th standard basis vector (applied only to the first 14 components of $\mathbf{x}$).

6.2 SVD-Based Rank Analysis

The Jacobian $\mathbf{J} \in \mathbb{R}^{N_r \times 14}$ is decomposed via SVD:

$$\mathbf{J} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^\top \tag{28}$$

where $\mathbf{\Sigma} = \mathrm{diag}(\sigma_1, \ldots, \sigma_{14})$ with $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_{14} \geq 0$.

The *observable rank* is the number of singular values exceeding a threshold $\sigma_{\min}$:

$$r = \left| \{k : \sigma_k > 1.5\sqrt{\lambda}\} \right| \tag{29}$$

The threshold $1.5\sqrt{\lambda}$ is chosen to be slightly above the Tikhonov regularization floor, so that parameters whose identifiability comes only from the regularization term are excluded.

6.3 Per-Parameter Observability Score

Each parameter is assigned an observability score based on its contribution to the well-observed singular directions:

$$s_j = \frac{1}{s_{\max}} \sum_{k:\, \sigma_k > \sigma_{\min}} \sigma_k^2 \cdot V_{k,j}^2 \tag{30}$$

where $s_{\max} = \max_j s_j$ normalizes the scores to $[0, 1]$. Parameters with $s_j > 0.05$ are marked as *well-observed* (V), while others are marked as *poorly observed* (X).

6.4 Interpretation for Single-Wall Calibration

With our 5×7 grid of captures facing a single flat wall, the analysis consistently identifies:

- **Observable rank: 7 out of 14.**
- Well-observed parameters: a subset of pan and laser translations.
- Poorly observed (null space): all pan rotations, encoder zeros, laser z-translation, and all laser rotations.

The 7 unobservable directions correspond to parameter combinations that produce identical projected point patterns on a single flat surface. This is a fundamental geometric limitation, not a software issue.

7 Validation

Validation measures how well the calibrated kinematic chain explains held-out (or same-set) observations, compared to the uncalibrated chain.

7.1 Validation Procedure

Given a validation dataset of K captures and the optimal correction $\boldsymbol{\delta}^*$:

1. **Apply line quality filter** (same threshold as solver).
2. For each capture k, transform inlier points to base frame: $\mathbf{p}_j^{\text{base}} = \mathbf{T}_{\text{laser}}^{\text{base}}(\theta_p^{(k)}, \theta_t^{(k)}; \boldsymbol{\delta}^*) \cdot (\mathbf{p}_j^L, 0, 1)^\top$
3. Fit a global plane $(\hat{\mathbf{n}}, d)$ via SVD (Section 5.2).
4. Compute per-capture plane RMSE:

$$\text{RMSE}_k = \sqrt{\frac{1}{|\mathcal{I}_k|} \sum_{j \in \mathcal{I}_k} (\hat{\mathbf{n}}^\top \mathbf{p}_j^{\text{base}} - d)^2} \times 1000 \qquad [\text{mm}] \tag{31}$$

5. Repeat steps 2–4 with $\boldsymbol{\delta} = \mathbf{0}$ (uncalibrated) for comparison.

7.2 Reported Metrics

- **Global RMSE:** $\text{RMSE}_{\text{global}} = \sqrt{\frac{1}{N}\sum_{i=1}^N e_i^2}$ where e_i are all point-to-plane signed distances across all captures.
- **Per-capture RMSE mean / std / max.**
- **Max point error:** $\max_i |e_i|$.
- **Improvement:** $(1 - \text{RMSE}_{\text{cal}}/\text{RMSE}_{\text{uncal}}) \times 100\%$.
- **PASS/FAIL verdict:** Based on whether the global RMSE falls below a user-defined threshold (default: 10 mm).

Important: The line quality filter must be applied consistently in both the solver and validation. Applying it only in the solver but evaluating on unfiltered data leads to misleading results (calibrated appearing worse than uncalibrated) because defect corrections amplify noise in poor-quality captures.

8 Experimental Results

8.1 Dataset

Table 2: Calibration dataset characteristics.

Property	Value
Total captures	31
Pan range	$-25°$ to $+25°$
Tilt range	$-15°$ to $+15°$
FOV half-angle	$10°$
Total inlier points	1612
After line filter (3 mm)	19 captures, 1012 points
Subsampled for solver	570 points

8.2 Euler Convention Bug

The most impactful finding was the Euler convention mismatch. Table 3 shows the effect of correcting from intrinsic `'xyz'` to extrinsic `'XYZ'`:

Table 3: Impact of correcting the Euler convention.

	`'xyz'` (intrinsic, wrong)	`'XYZ'` (extrinsic, correct)
Uncalibrated plane RMSE	56 mm	27 mm

8.3 Line Quality Filter Impact

Table 4: Effect of line quality filter on uncalibrated RMSE.

	Unfiltered (31 cap)	Filtered (19 cap)
Uncalibrated plane RMSE	26.9 mm	16.2 mm
Reduction	—	40%

8.4 Solver and Validation Results

Table 5: Calibration results with different solver settings (all with 3 mm line filter).

Settings	λ	B_t / B_r / B_e	Obs. Rank	Cost ↓	Global RMSE ↓
Tight bounds	0.1	5 mm / 2.87° / 3°	7/14	20.1%	10.1%
Wide encoder	0.1	5 mm / 2.87° / 10°	7/14	22.3%	12.3%
Strong reg.	1.0	5 mm / 2.87° / 10°	6/14	21.4%	12.3%

The best configuration achieves a **12.3% reduction** in global plane RMSE (16.24 mm → 14.24 mm) with 15 of 19 captures improving individually.

8.5 Validation Summary (Best Configuration)

Table 6: Validation statistics for the best configuration ($\lambda = 0.1$, $B_t = 5$ mm, $B_r = 2.87°$, $B_e = 10°$).

Metric	Uncalibrated	Calibrated	Improvement
Global RMSE (mm)	16.24	14.24	+12.3%
Per-capture RMSE mean	14.44	12.52	+13.3%
Per-capture RMSE std	7.50	6.83	+8.9%
Per-capture RMSE max	37.82	33.65	+11.0%
Max point error (mm)	44.54	44.49	+0.1%

8.6 Parameters at Bounds

Seven of the 14 parameters consistently hit their bounds, indicating the solver desires corrections larger than what we permit:

- `enc_tilt_zero`: $-10.0°$ (at bound)
- `dt_laser_x`: -5.0 mm (at bound)
- `dt_laser_y`: $+5.0$ mm (at bound)
- `dt_laser_z`: $+5.0$ mm (at bound)
- `dr_laser_roll`, `_pitch`, `_yaw`: all at $\pm 2.87°$

The enc_tilt_zero absorbs whatever bound is given, which is a hallmark of a degenerate parameter—it trades off with laser rotation corrections.

9 Limitations and Future Work

9.1 Fundamental Limitation: Single-Wall Observability

The calibration achieves only 7/14 observable parameters because a single flat wall provides limited geometric diversity. Multiple scan configurations viewing the same plane can only constrain parameters that change the *relative* arrangement of points on that plane. Parameters that rigidly translate or rotate all points together (such as tilt encoder offset vs. laser rotation about the tilt axis) are perfectly confounded.

9.2 Recommendations for Improved Calibration

To break the degeneracy and improve calibration accuracy, the following strategies should be pursued:

1. **Multiple walls at different orientations.** Capture data facing 2–3 walls at different angles. The different surface normals provide linearly independent constraints that disambiguate currently degenerate parameters. A room corner (two perpendicular walls) is the simplest improvement.

2. **Planar targets at different distances.** Walls at varying distances improve sensitivity to translation parameters (which scale linearly with range).

3. **Geometric features.** A corner, edge, or V-shaped target provides line-like constraints that constrain both the position and orientation of the laser frame simultaneously.

4. **Wider joint-space coverage.** The current $\pm 25°$ pan / $\pm 15°$ tilt range is limited. If the mechanism allows, sweeping over a wider range increases sensitivity to axis alignment errors.

5. **Higher-quality lidar sensor.** The Hokuyo URG-04LX has specification accuracy of ± 60 mm at ranges beyond 1 m and inherent noise of 2–5 mm even at close range. A higher-precision sensor (e.g., UTM-30LX) would lower the noise floor.

6. **Repeated scans and averaging.** Taking multiple scans at each configuration and averaging reduces random noise. This is a simple change requiring only extra acquisition time.

7. **Model-based reflectance filtering.** The laser's signal quality varies with incidence angle and surface material. Filtering by signal intensity (if available) could further improve data quality.

8. **Reduce the model.** Given that only 7 parameters are observable, one could fix the remaining 7 at zero and re-solve with only the identifiable subset. This eliminates regularization bias on the observable parameters. The risk is that the wrong parameters may be selected; the SVD analysis should guide this selection.

9. **Iterative calibration.** After an initial calibration updates the URDF, re-collect data and solve again. The residual errors after the first round may reveal previously masked effects, allowing iterative refinement.

9.3 Summary of Achievable Accuracy

Table 7: Expected accuracy vs. calibration environment complexity.

Environment	Expected Obs. Rank	Estimated RMSE
Single wall (current)	7/14	14–16 mm
Two walls (corner)	10–12/14	5–8 mm
Three walls + varying distances	13–14/14	2–5 mm
High-precision lidar + multi-wall	14/14	<2 mm

10 Conclusion

We presented a complete kinematic calibration pipeline for a pan-tilt mounted Hokuyo URG-04LX planar LiDAR system. The pipeline includes automated grid-scan data acquisition, RANSAC-based line fitting, per-capture line quality filtering, 14-parameter kinematic correction estimation via Levenberg–Marquardt optimization, SVD-based observability analysis, and comprehensive validation.

Three key findings emerged:

1. A single-character Euler convention bug (`'xyz'` → `'XYZ'`) was responsible for 50% of the initial error, highlighting the critical importance of matching rotation conventions between the URDF parser and the solver.
2. The per-capture line quality filter—checking whether individual scans conform to a straight line before including them in the plane fit—produced a 40% noise reduction, the single largest improvement in the pipeline.
3. Single-wall calibration fundamentally limits observability to 7 of 14 parameters, achieving 12.3% RMSE improvement. Breaking through this ceiling requires geometric diversity in the calibration environment.

The complete pipeline is implemented as a ROS package (`ptu_lidar_calib`) with modular scripts for acquisition, solving, and validation, ready for deployment on multi-wall environments.

A Complete Forward Kinematics Derivation

We detail the full matrix chain for the uncorrected ($\boldsymbol{\delta} = \mathbf{0}$) case. From the URDF:

Step 1: Base to Pan Link.

$$\mathbf{T}_{\text{pan}}^{\text{nom}} = \begin{bmatrix} \mathbf{I}_3 & (0,\ -0.0285,\ 0.092)^\top \\ \mathbf{0}^\top & 1 \end{bmatrix} \tag{32}$$

(Identity rotation since RPY $= (0, 0, 0)$.)

Step 2: Pan Rotation. Joint axis is $(0, 0, -1)$, so for commanded angle θ_p:

$$\mathbf{R}_{\text{pan}}(\theta_p) = \mathbf{R}_z(-\theta_p) = \begin{bmatrix} \cos\theta_p & \sin\theta_p & 0 \\ -\sin\theta_p & \cos\theta_p & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{33}$$

Step 3: Pan Link to Tilt Link.

$$\mathbf{T}_{\text{tilt}}^{\text{nom}} = \mathbf{T}\Big((0,\ -0.009,\ 0.022),\ (0,\ 1.57,\ 0)\Big) \tag{34}$$

The RPY (0, 1.57, 0) applies a $\approx 90°$ pitch rotation:

$$\mathbf{R}_y(1.57) \approx \begin{bmatrix} \cos 1.57 & 0 & \sin 1.57 \\ 0 & 1 & 0 \\ -\sin 1.57 & 0 & \cos 1.57 \end{bmatrix} \approx \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{bmatrix} \tag{35}$$

This rotates the local z-axis to align with the global x-axis, so the tilt joint's rotation about local z becomes a rotation about the (approximate) global x-axis—i.e., a tilt.

Step 4: Tilt Rotation. Joint axis is $(0, 0, 1)$, so: $\mathbf{R}_{\text{tilt}}(\theta_t) = \mathbf{R}_z(\theta_t)$.

Step 5: Tilt Link to Laser. The composite of two fixed joints:

$$\mathbf{T}_{\text{laser}}^{\text{nom}} = \underbrace{\mathbf{T}\Big((-0.045, 0, 0),\ (0, 0, 0)\Big)}_{\text{hokuyo_base_joint}} \cdot \underbrace{\mathbf{T}\Big((-0.061, 0.014, 0.040),\ (1.57, 3.14, -1.57)\Big)}_{\text{hokuyo_joint}} \tag{36}$$

Full Chain.

$$\mathbf{T}_{\text{laser}}^{\text{base}} = \mathbf{T}_{\text{pan}}^{\text{nom}} \cdot \mathbf{R}_z(-\theta_p) \cdot \mathbf{T}_{\text{tilt}}^{\text{nom}} \cdot \mathbf{R}_z(\theta_t) \cdot \mathbf{T}_{\text{laser}}^{\text{nom}} \tag{37}$$

B Parameter Indices

Table 8: Mapping from parameter index to physical meaning.

Index	Label	Meaning	Unit
0	`dt_pan_x`	Pan origin x translation	m
1	`dt_pan_y`	Pan origin y translation	m
2	`dt_pan_z`	Pan origin z translation	m
3	`dr_pan_roll`	Pan origin roll correction	rad
4	`dr_pan_pitch`	Pan origin pitch correction	rad
5	`dr_pan_yaw`	Pan origin yaw correction	rad
6	`enc_pan_zero`	Pan encoder zero offset	rad
7	`enc_tilt_zero`	Tilt encoder zero offset	rad
8	`dt_laser_x`	Laser mount x translation	m
9	`dt_laser_y`	Laser mount y translation	m
10	`dt_laser_z`	Laser mount z translation	m
11	`dr_laser_roll`	Laser mount roll correction	rad
12	`dr_laser_pitch`	Laser mount pitch correction	rad
13	`dr_laser_yaw`	Laser mount yaw correction	rad